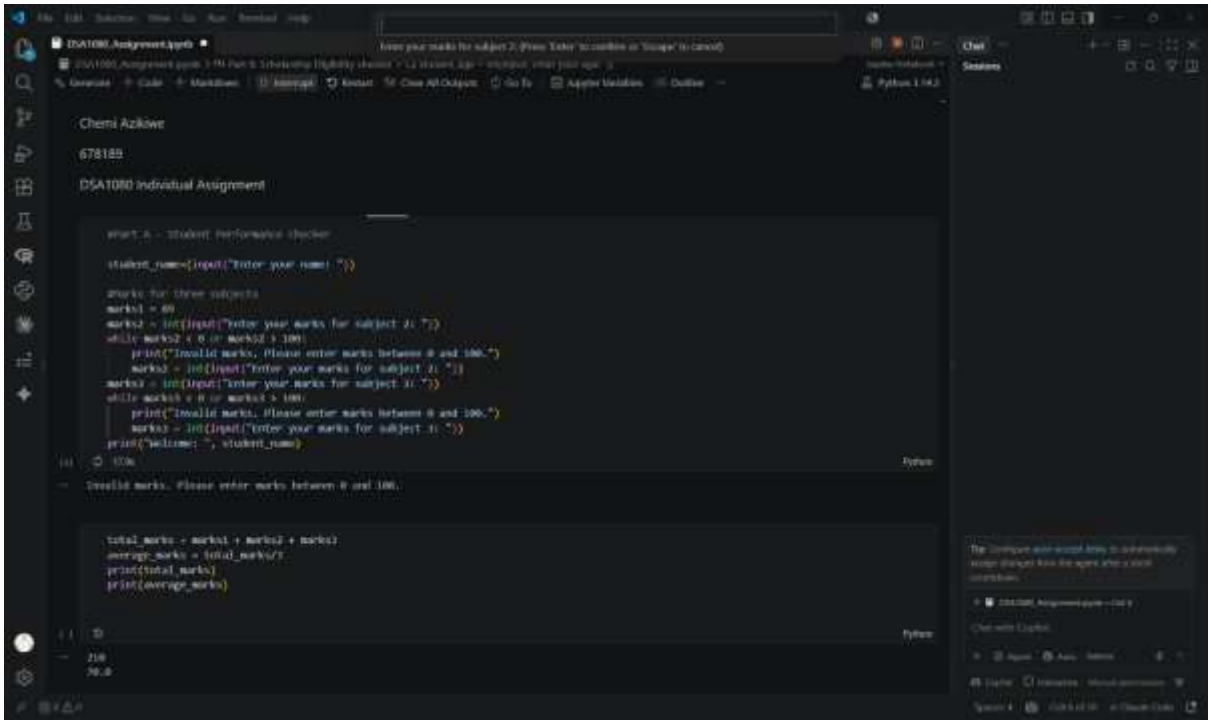
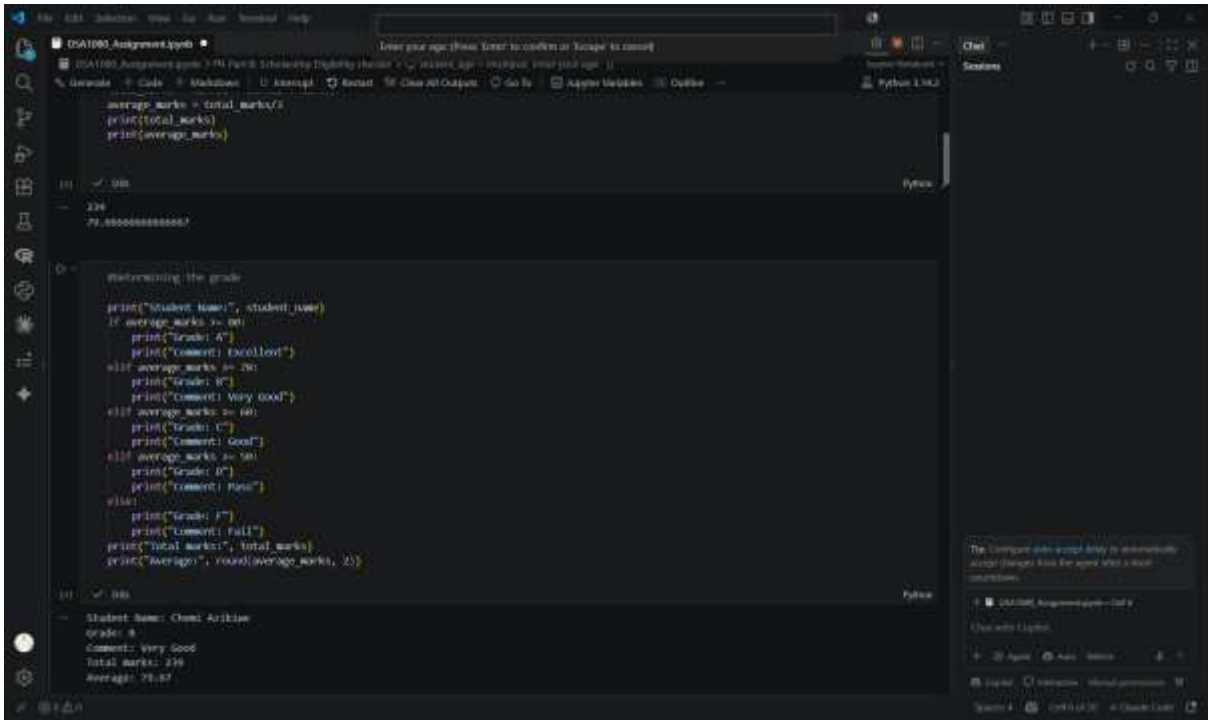
Part A

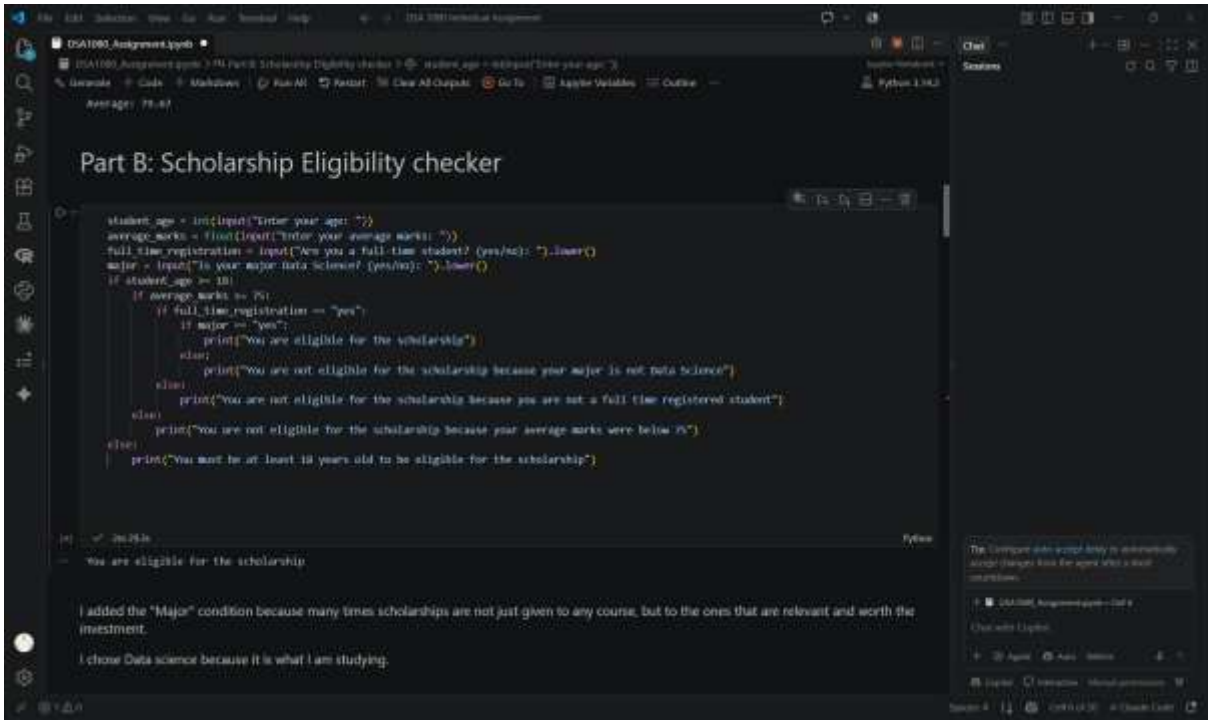
Invalid Run



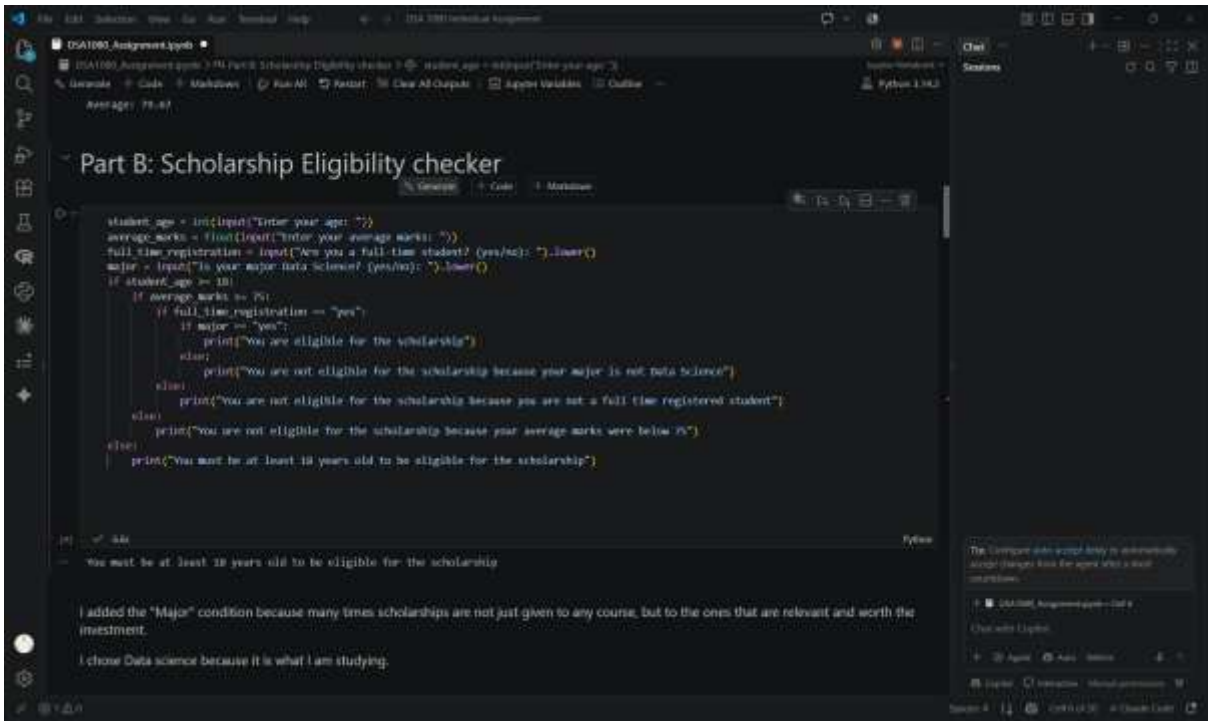
Valid run



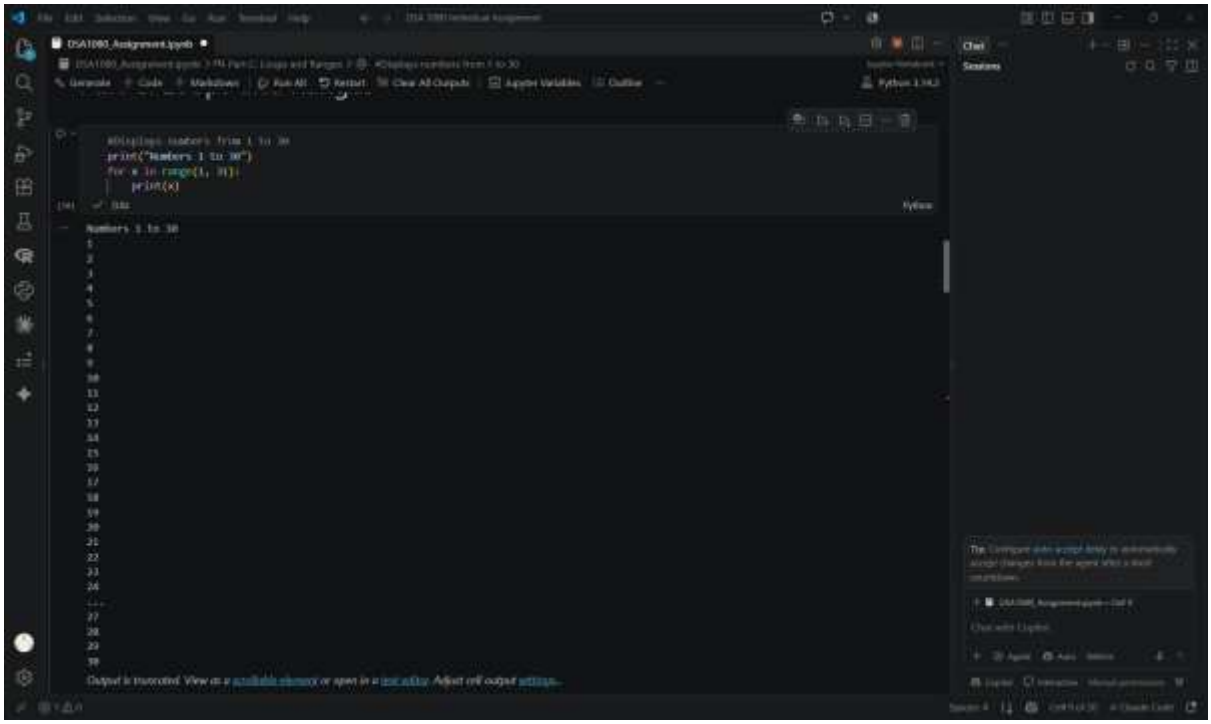
Part B Scholarship Eligibility Checker

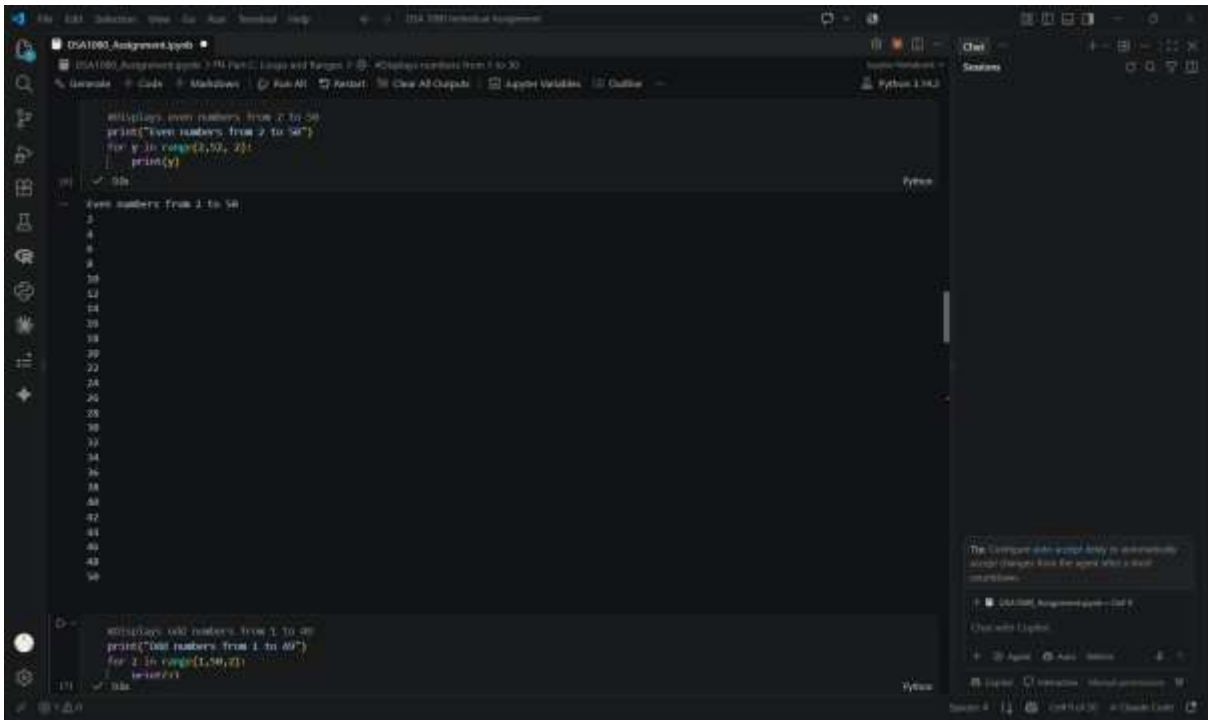


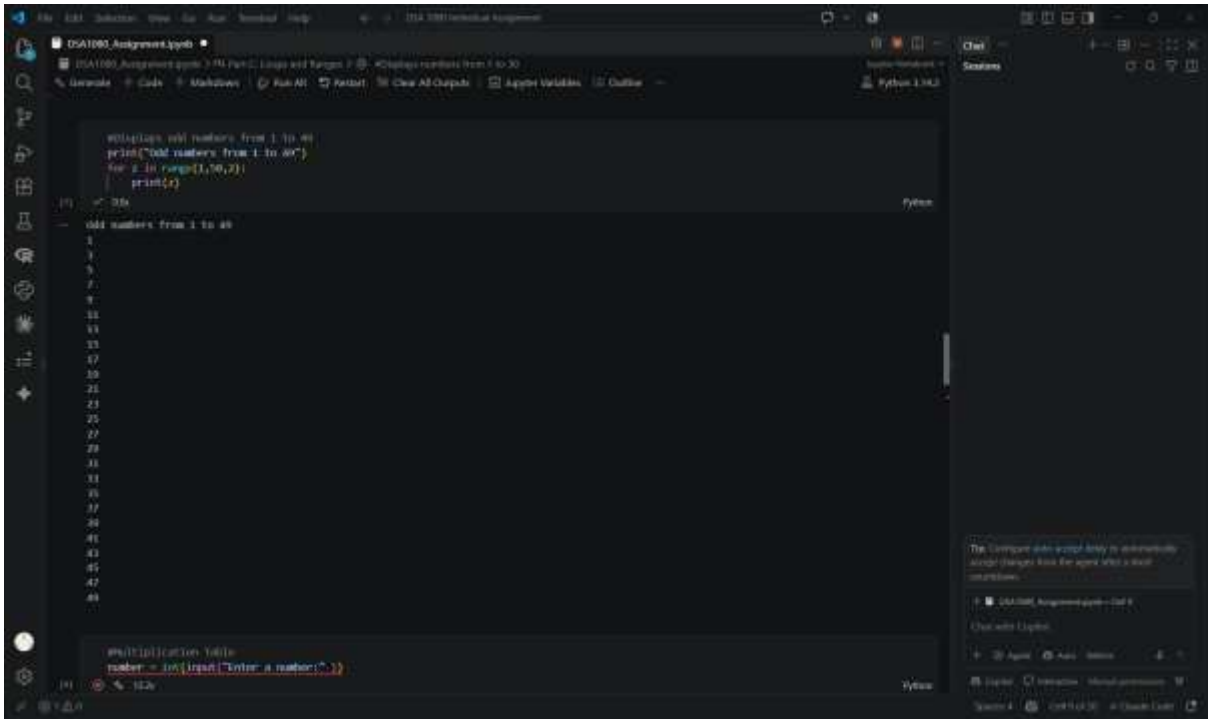
Re-run when age is too low



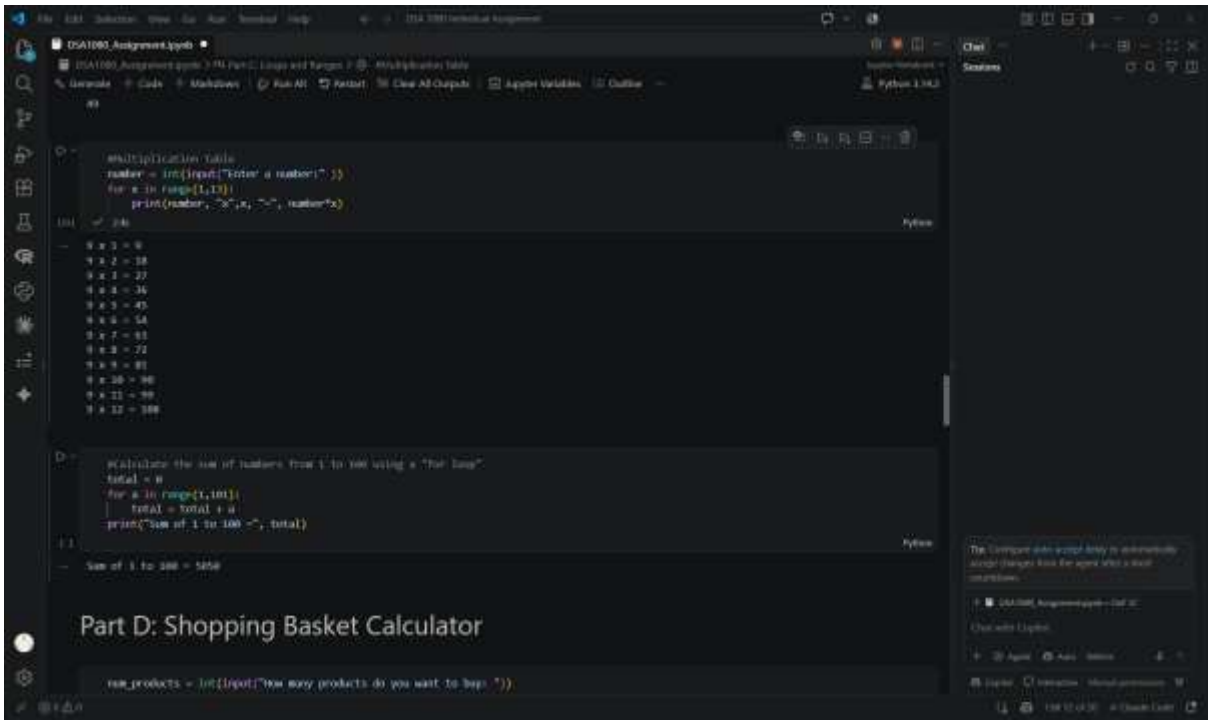
Part C


DSA1080_Assignment.ipynb
#Displays numbers from 1 to 30
print("Numbers 1 to 30")
for x in range(1, 31):
print(x)
Numbers 1 to 30
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
...
27
28
29
30
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
Chat
Sessions

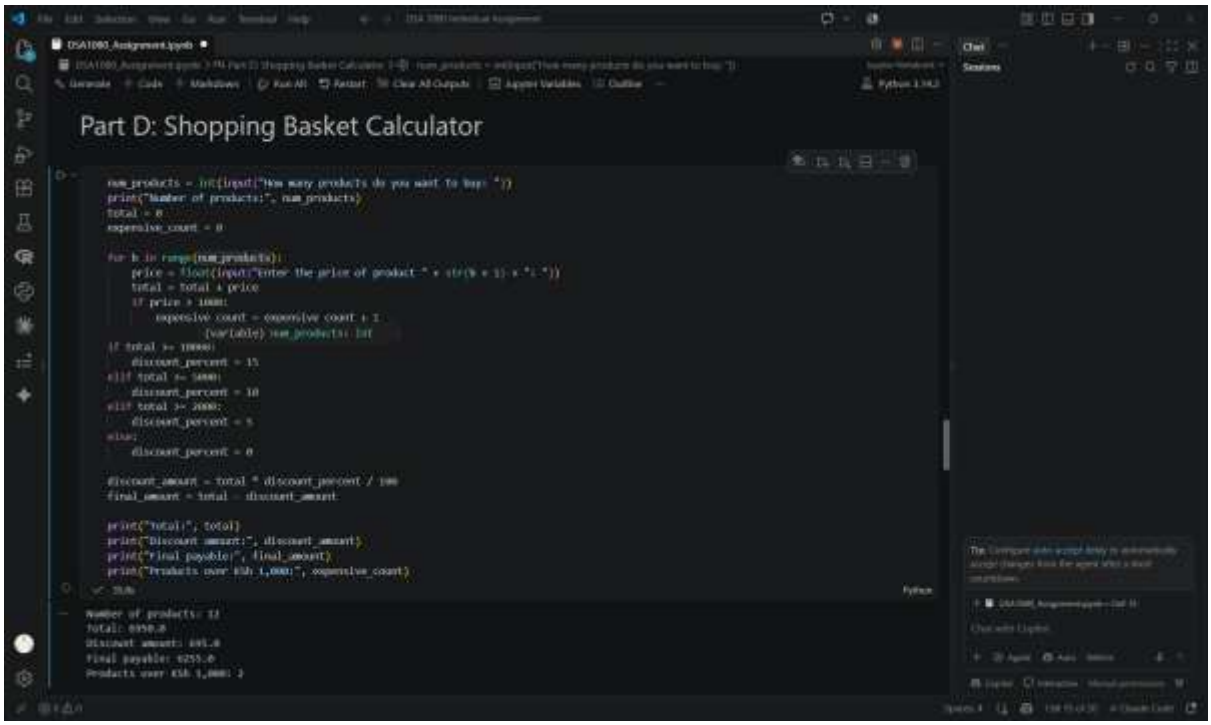

DSA1080_Assignment.ipynb
#Displays even numbers from 2 to 50
print("Even numbers from 2 to 50")
for y in range(2,52, 2):
print(y)
Even numbers from 2 to 50
2
4
6
8
10
12
14
16
18
20
22
24
26
28
30
32
34
36
38
40
42
44
46
48
50
#Displays odd numbers from 1 to 49
print("Odd numbers from 1 to 49")
for i in range(1,50,2):
print(i)
Chat
Sessions



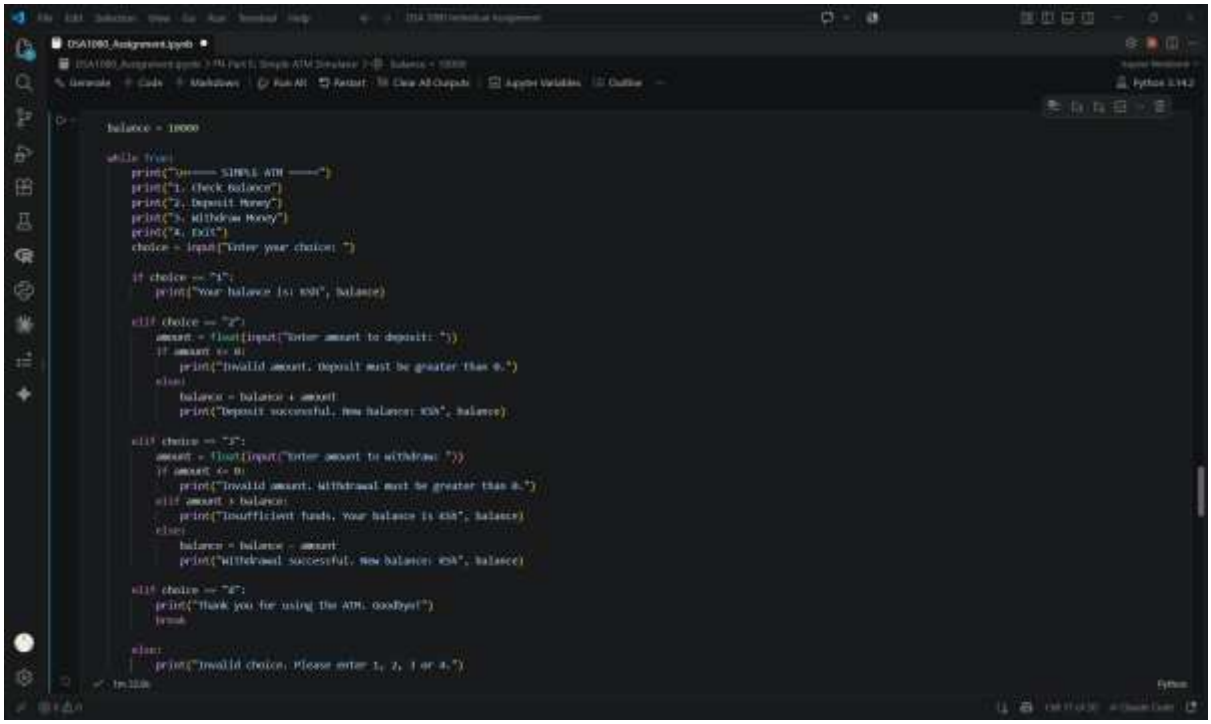
Multiplication table



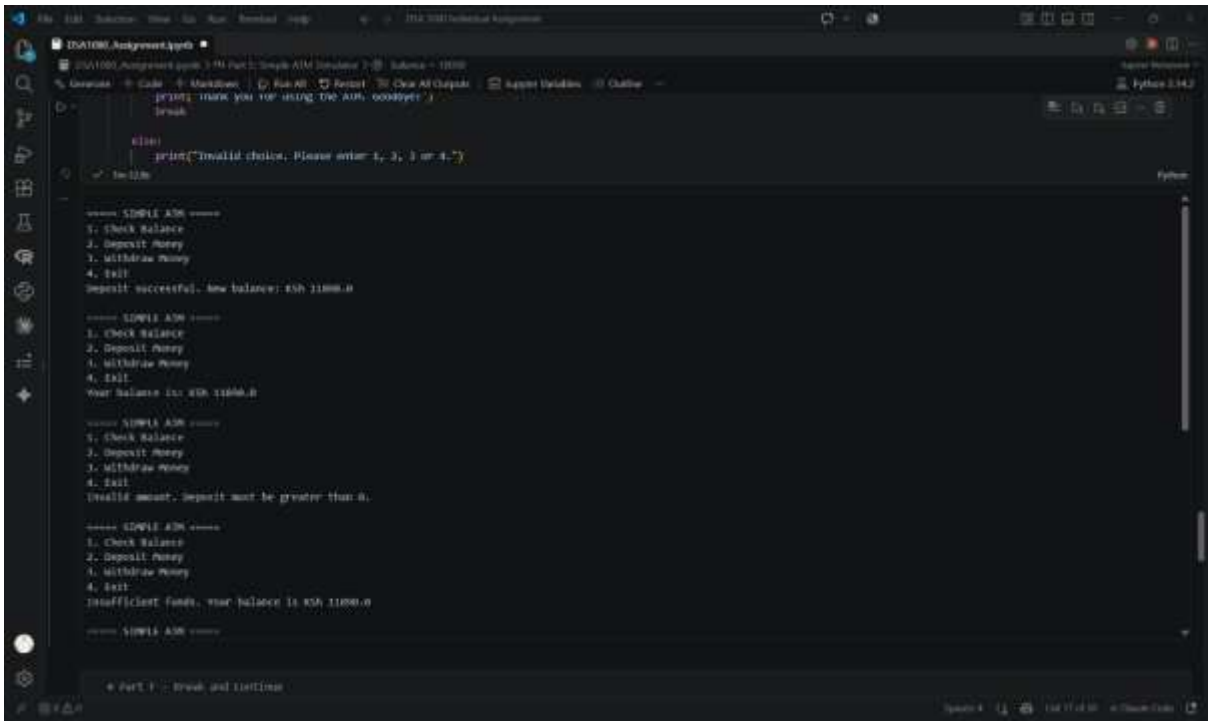
Part D Shopping Basket Calculator

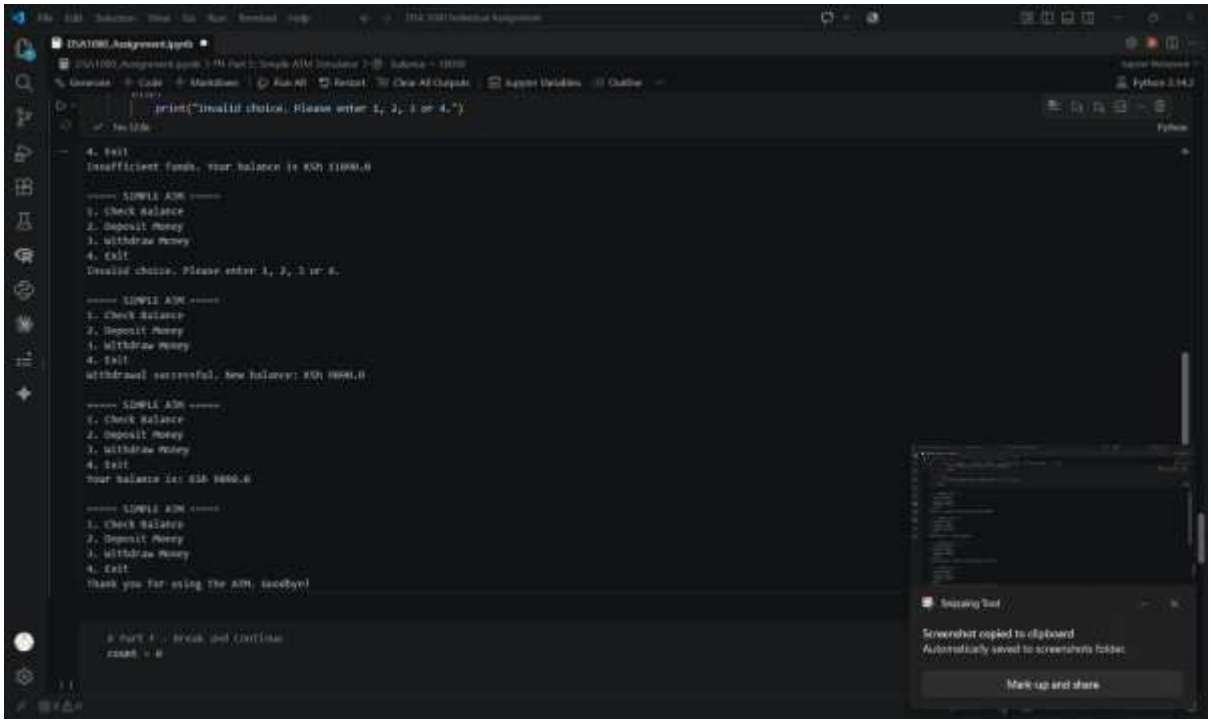


Part E Simple ATM calculator

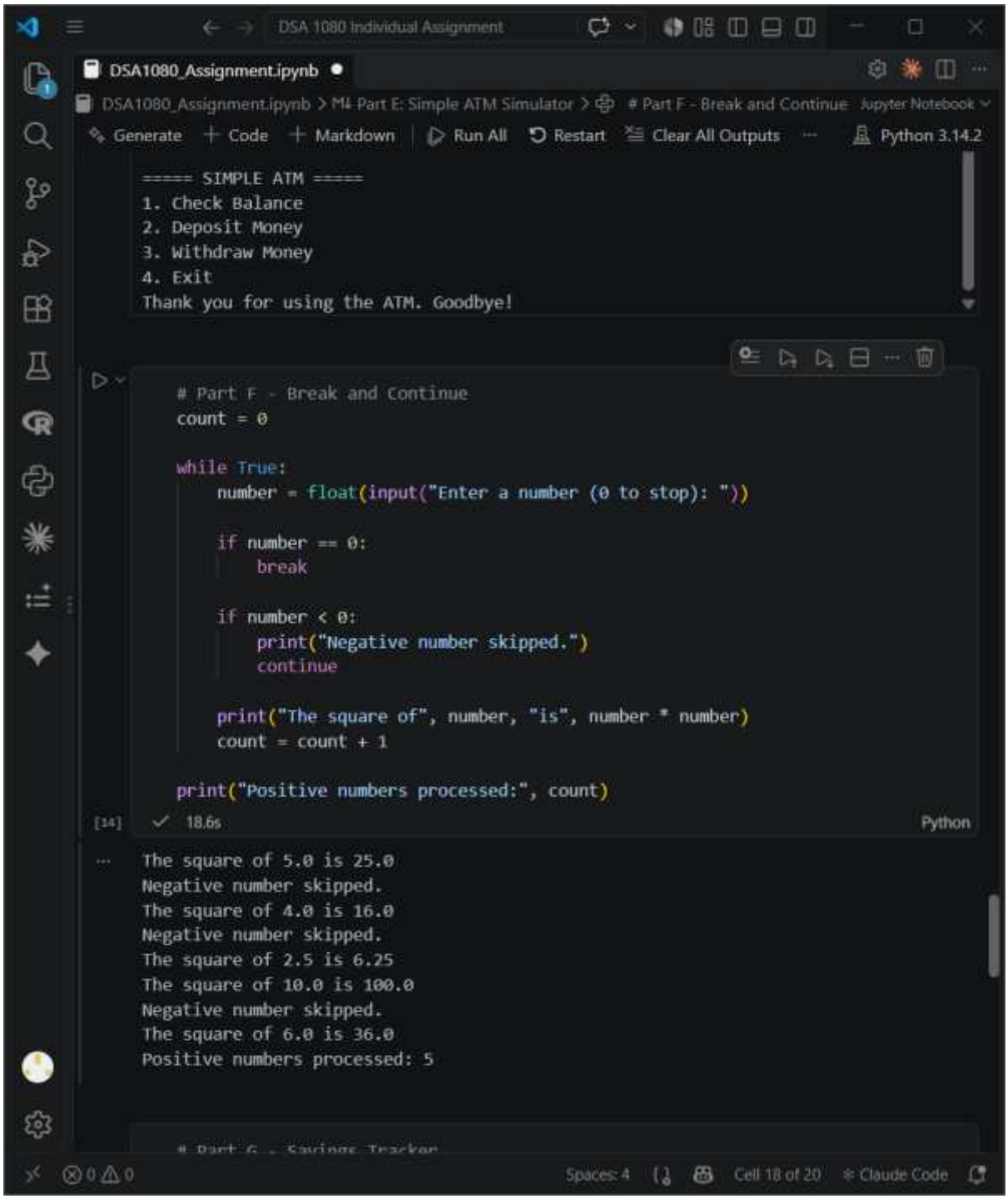


Output

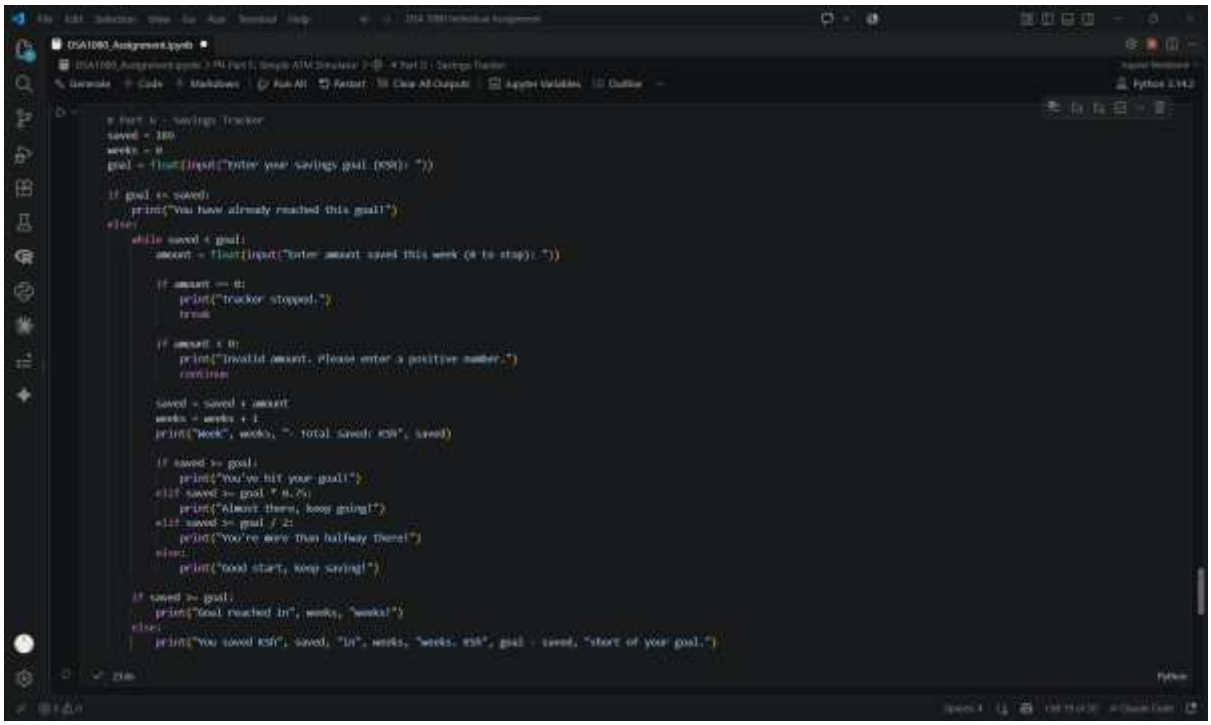

print("Invalid choice. Please enter 1, 2, 3 or 4.")
===== SIMPLE ATM =====
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Deposit successful. New balance: KSh 11000.0
===== SIMPLE ATM =====
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Your balance is: KSh 11000.0
===== SIMPLE ATM =====
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Invalid amount. Deposit must be greater than 0.
===== SIMPLE ATM =====
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Insufficient funds. Your balance is KSh 11000.0
===== SIMPLE ATM =====


print("Invalid choice. Please enter 1, 2, 3 or 4.")
4. Exit
Insufficient funds. Your balance is KSh 11000.0
===== SIMPLE ATM =====
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Invalid choice. Please enter 1, 2, 3 or 4.
===== SIMPLE ATM =====
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Withdrawal successful. New balance: KSh 8000.0
===== SIMPLE ATM =====
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Your balance is: KSh 8000.0
===== SIMPLE ATM =====
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
Thank you for using the ATM. Goodbye!
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Mark up and share

Part F Break and Continue



Part G Savings Tracker



```
# Part G - Savings Tracker
saved = 189
weeks = 0
goal = float(input("Enter your savings goal (KSh): "))

if goal <= saved:
    print("You have already reached this goal!")
else:
    while saved < goal:
        amount = float(input("Enter amount saved this week (0 to stop): "))

        if amount == 0:
            print("Tracker stopped.")
            break

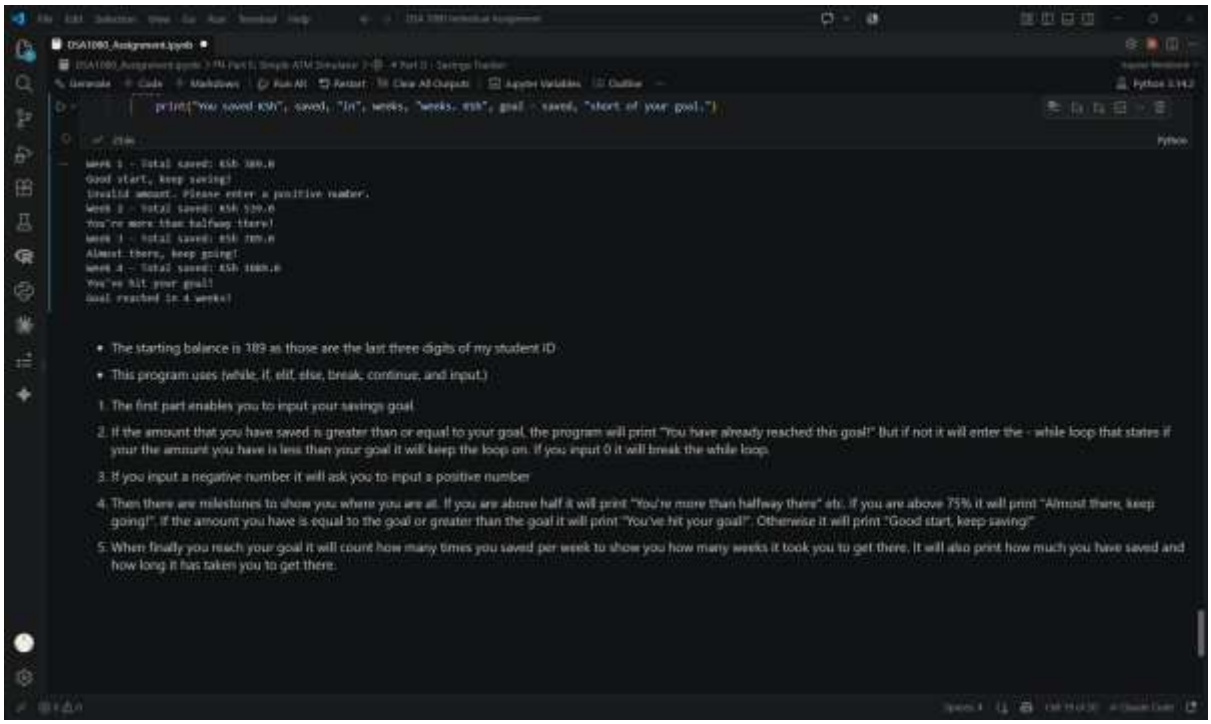
        if amount < 0:
            print("Invalid amount. Please enter a positive number.")
            continue

        saved = saved + amount
        weeks = weeks + 1
        print("Week", weeks, "- Total saved: KSh", saved)

        if saved >= goal:
            print("You've hit your goal!")
        elif saved >= goal * 0.75:
            print("Almost there, keep going!")
        elif saved >= goal / 2:
            print("You're more than halfway there!")
        else:
            print("Good start, Keep saving!")

    if saved >= goal:
        print("Goal reached in", weeks, "weeks!")
    else:
        print("You saved KSh", saved, "in", weeks, "weeks. KSh", goal - saved, "short of your goal.")
```

Output



```
Week 1 - Total saved: KSh 389.0
Good start, Keep saving!
Invalid amount. Please enter a positive number.
Week 2 - Total saved: KSh 539.0
You're more than halfway there!
Week 3 - Total saved: KSh 789.0
Almost there, keep going!
Week 4 - Total saved: KSh 1089.0
You've hit your goal!
Goal reached in 4 weeks!
```

- The starting balance is 189 as those are the last three digits of my student ID
- This program uses (while, if, elif, else, break, continue, and input.)

1. The first part enables you to input your savings goal.
2. If the amount that you have saved is greater than or equal to your goal, the program will print "You have already reached this goal!" But if not it will enter the - while loop that states if your the amount you have is less than your goal it will keep the loop on. If you input 0 it will break the while loop.
3. If you input a negative number it will ask you to input a positive number
4. Then there are milestones to show you where you are at. If you are above half it will print "You're more than halfway there" etc. If you are above 75% it will print "Almost there, keep going!", if the amount you have is equal to the goal or greater than the goal it will print "You've hit your goal!". Otherwise it will print "Good start, keep saving!"
5. When finally you reach your goal it will count how many times you saved per week to show you how many weeks it took you to get there. It will also print how much you have saved and how long it has taken you to get there.